UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 21

Content.

1. Rademacher / Gaussian averages.
2. Gaussian comparison inequalities.

1 Rademacher Averages

1.1 Setup

Let $\mathcal{F} = \{f : X \to \mathbb{R}\}$ be a class of functions. We want to bound the excess risk, which requires controlling

$$\mathbb{E}_X \left[\sup_{f \in \mathcal{F}} \left| \mathbb{E}[f(X)] - \frac{1}{n} \sum_{i=1}^{n} f(X_i) \right| \right]. \quad (\star)$$

The symmetrization lemma gives

$$(\star) \leq 2\, \mathbb{E}_{\epsilon, X} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \epsilon_i f(X_i) \right| \right],$$

where ϵ_i are i.i.d. Rademacher r.v.s. Condition on data $(X_1, ..., X_n) = (x_1, ..., x_n)$, we want to bound

$$\frac{1}{\sqrt{n}} \mathbb{E}_\epsilon \left[\sup_{f \in \mathcal{F}} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \epsilon_i f(x_i) \right].$$

Let us define

$$\Theta = \left\{ \frac{1}{\sqrt{n}} (f(x_1), f(x_2), ..., f(x_n)),\ f \in \mathcal{F} \right\}.$$

For $\theta = \frac{1}{\sqrt{n}}(f(x_1), \ldots, f(x_n)) \in \Theta$ and $\theta' = \frac{1}{\sqrt{n}}(f'(x_1), \ldots, f'(x_n)) \in \Theta$,

$$d(\theta, \theta')^2 = \|f - f'\|_n^2 = \frac{1}{n}\sum_{i=1}^{n}(f(x_i) - f'(x_i))^2.$$

Remark. $\|f - f'\|_n^2$ is a function of data $x_1, ..., x_n$ and it is a pesudometric of $\mathcal{F}$.

1.2 Bounding the Rademacher Process

We observe:

$$\mathbb{E}_\epsilon\left[\sup_{f\in\mathcal{F}}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\epsilon_i f(x_i)\right] = \mathbb{E}_\epsilon\left[\sup_{\theta\in\Theta}\langle\theta, \epsilon\rangle\right].$$

And

$$\mathcal{N}(\Theta, \|\|_2, \delta) = \mathcal{N}(\mathcal{F}, \|\|_n, \delta).$$

By Dudley's chaining bound, for a sub-Gaussian process gives,

$$\mathbb{E}\left[\sup_{\theta\in\Theta}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\epsilon_i\theta_i\right] \le 2\delta\sqrt{n} + 8\sqrt{n}\int_{\delta/4}^{D/2}\sqrt{\log\mathcal{N}(\mathcal{F}, \|\|_n, \epsilon)}\,d\epsilon,$$

where $\mathcal{N}(\Theta, \|\|_n, \epsilon)$ is the covering number, and D is the diameter.

Corollary 1. For any $x_1, ..., x_n$,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}\epsilon_i f(x_i)\right] \le \inf_{\delta>0}\left\{8\delta + \frac{12}{\sqrt{n}}\int_{\delta}^{D/2}\sqrt{\log\mathcal{N}(\mathcal{F}, \|\|_n, \epsilon)}d\epsilon\right\}$$

So symmetrization lemma gives

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\mathbb{E}f(x) - \frac{1}{n}\sum_{i=1}^{n}f(x_i)\right] \le \mathbb{E}\left[\inf_{\delta>0}\left\{16\delta + \frac{24}{\sqrt{n}}\int_{\delta}^{D/2}\sqrt{\log\mathcal{N}(\mathcal{F}, \|\|_n, \epsilon)}d\epsilon\right\}\right]$$

where $D = \sup_{f\in\mathcal{F}}\|f\|_n = \sqrt{\sum_i f(x_i)^2/n} \le \sup_{f\in\mathcal{F}}\|f\|_\infty$.

2 Gaussian Comparison Inequalities

2.1 Setup

Consider two centered sub-Gaussian processes

$$\{X_t,\ t \in T\}, \qquad \{Y_t,\ t \in T\},$$

where (T, d) a metric space. We want to compare them. For a function F, how does $F(X)$ relate to $F(Y)$? One way is to compare

$$F(X) = \mathbb{E}\left[\sup_{t\in T} X_t\right] \quad \text{vs.} \quad F(Y) = \mathbb{E}\left[\sup_{t\in T} Y_t\right].$$

2.2 Gaussian Comparison Theorem

Theorem 1. Let $(X_1, \ldots, X_N)$ and $(Y_1, \ldots, Y_N)$ be a pair of centered sub-Gaussian random vectors. Let $A, B \subset [N] \times [N]$ be disjoint subsets and suppose:

- $\mathbb{E}[X_i X_j] \le \mathbb{E}[Y_i Y_j]$ for $(i, j) \in A$,
- $\mathbb{E}[X_i X_j] \ge \mathbb{E}[Y_i Y_j]$ for $(i, j) \in B$,
- $\mathbb{E}[X_i X_j] = \mathbb{E}[Y_i Y_j]$ otherwise.

Let F be a twice differentiable function with

$$\frac{\partial^2 F}{\partial u_i \partial u_j} \ge 0 \quad \text{for } (i, j) \in A, \qquad \frac{\partial^2 F}{\partial u_i \partial u_j} \le 0 \quad \text{for } (i, j) \in B.$$

Then

$$\mathbb{E}F(X) \le \mathbb{E}F(Y).$$

Proof. Assume $X = (X_1, ..., X_N) \perp Y = (Y_1, ..., Y_N)$. Denote the interpolation function $Z(t) = \sqrt{1-t}X + \sqrt{t}Y \in \mathbb{R}^n, t \in [0, 1]$. Define $\phi(t) = \mathbb{E}F(Z(t))$. Two special cases are

$$\phi(0) = \mathbb{E}F(X), \quad \phi(1) = \mathbb{E}F(Y).$$

If $\phi'(t) \ge 0, \forall t \in [0, 1]$, we directly show $\mathbb{E}F(Y) \ge \mathbb{E}F(X)$. By chain rule,

$$\phi'(t) = \sum_{j=1}^{N} \mathbb{E}\left[\frac{\partial F}{\partial Z_j}(Z(t)) Z_j'(t)\right]$$

and express $Z'j(t)$ as:

$$Z_j'(t) = \frac{d}{dt}(\sqrt{1-t}X_j + \sqrt{t}Y_j) = -\frac{1}{\sqrt{1-t}}X_j + \frac{1}{2\sqrt{t}}Y_j.$$

Moreover,

$$\begin{aligned}\mathbb{E}[Z_i(t)Z_j'(t)] &= \mathbb{E}[\sqrt{1-t}X_i + \sqrt{t}Y_i]\mathbb{E}\left[-\frac{1}{\sqrt{1-t}}X_j + \frac{1}{2\sqrt{t}}Y_j\right] \\ &= \frac{1}{2}(\mathbb{E}Y_iY_j - \mathbb{E}X_iX_j).\end{aligned}$$

Next, for fixed j and $1 \leq i \leq N$, we write $Z_i(t) = \alpha_{ij}Z_j'(t) + w_{ij}$, with

$$\alpha_{ij}\begin{cases}\geq 0, \text{ if } (i,j) \in A \\ \leq 0, \text{ if } (i,j) \in B \\ = 0, \text{ otherwise}\end{cases}$$

and $w_{ij} \perp z_j'(t)$. So $w(j) = \begin{pmatrix} w_{1j} \\ ... \\ w_{Nj}\end{pmatrix}$ is independent of $Z_j'(t)$. First order Taylor expansion of $\frac{\partial F}{\partial Z_j}$ between $Z(t)$ and $w(j)$ gives:

$$\frac{\partial F}{\partial Z_j}(Z(t)) = \frac{\partial F}{\partial Z_j}(w(j)) + \sum_{i=1}^{N}\frac{\partial^2 F}{\partial Z_j Z_i}(u)\alpha_{ij}Z_j'(t)$$

for some $u \in \mathbb{R}^n$ between $w(j)$ and $Z(t)$. Thus,

$$\mathbb{E}\left[\frac{\partial F}{\partial Z_j}(Z(t))Z_j'(t)\right] = \underbrace{\mathbb{E}\left[\frac{\partial F}{\partial Z_j}(w(j))Z_j'(t)\right]}_{=0} + \mathbb{E}\left[\sum_{i=1}^{N}\underbrace{\frac{\partial^2 F}{\partial Z_j Z_i}(u)\alpha_{ij}Z_j'(t)^2}_{\geq 0}\right] \geq 0.$$

Thus $\phi(t)' \geq 0$ and ϕ is non-decreasing function, We arrive at

$$\mathbb{E}F(Y) = \phi(1) \geq \phi(0) = \mathbb{E}F(X)$$

□

2.3 Slepian's Inequality

Corollary 2 (Slepian's Inequality)**.** Let $X, Y \in \mathbb{R}^N$ be zero-mean Gaussian vectors with

$$\begin{aligned} &\mathbb{E}X_i^2 = \mathbb{E}Y_i^2, && \forall i \\ &\mathbb{E}[X_i X_j] \geq \mathbb{E}[Y_i Y_j] && \forall i \neq j. \end{aligned}$$

Then

$$\mathbb{E}\max_i X_i \ \leq \ \mathbb{E}\max_i Y_i.$$

An examples: $X = (Z, ..., Z)$ with $Z \sim \mathcal{N}(0,1)$ and $Y = (Y_1, ..., Y_N)$ with $Y_i \sim \mathcal{N}(0,1)$. X has entries more correlated than Y and $\mathbb{E}[X] = 0 \leq \mathbb{E}[Y] \leq \sqrt{2\log N}$.

Proof. Define the function $F_\beta(X) : \mathbb{R}^N \to \mathbb{R}$:

$$F_\beta(X) = \frac{1}{\beta}\log\sum_{j=1}^N e^{\beta X_j}$$

A lower bound of $F_\beta(X)$ is

$$F_\beta(X) = \frac{1}{\beta}\log\sum_{i=1}^N e^{\beta X_j} \geq \frac{1}{\beta}\log e^{\beta\max_j X_j} = \max_j X_j.$$

An upper bound of $F_\beta(X)$ is

$$F_\beta(x) \leq \frac{1}{\beta}\log\sum_{i=1}^N e^{\beta\max_j X_j} = \frac{1}{\beta}\log e^{N\beta\max_j X_j} = \frac{\log N}{\beta} + \max_j X_j$$

So

$$\max_j X_j \leq F_\beta(X) \leq \frac{\log N}{\beta} + \max_j X_j$$

in which $\beta > 0$ is the inverse temperature. As $\beta \to \infty$, $\lim_{\beta\to\infty} F_\beta(X) \to \max_j X_j$. Fact:

$$\frac{\partial F_\beta}{\partial x_i \partial x_j} \leq 0, \quad \forall i \neq j$$

Apply previous results with

$$A = \emptyset, \quad B = \{(i,j) : i \neq j\}$$

□

We have

$$\mathbb{E}F_\beta(X) \leq \mathbb{E}F_\beta(Y)$$
$$\mathbb{E}\max_j X_j \leq \mathbb{E}\max_j Y_j + \log\frac{N}{\beta}$$

Since $\beta > 0$ is arbitrary, taking $\beta \to \infty$ gives Slepian's result.